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clc;
clear;
close all;
% Generate input signal
x = randn(1,1000);
% Generate echo signal
d = filter([0.8 0.5 0.3], 1, x);
% Exponential error signals
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));
% Input and Echo Signal
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input', 'Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');
% LMS Output
figure;
plot(e1);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');
% NLMS Output
figure;
plot(e2);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');
% MSE Comparison
figure;
semilogy(e1.^2, 'r');
hold on;
semilogy(e2.^2, 'b');
grid on;
legend('LMS', 'NLMS');
title('MSE Comparison');
xlabel('Samples');
ylabel('Squared Error');

```

